

Renal Reference — BUN:Cr Ratio

BUN:Cr ratio tells you how much urea is in the blood relative to creatinine.

Normal = ~**10-15:1**. Prerenal = >20:1 (kidneys concentrate urea avidly when perfusion is low).

BUN rises when:

- Urea production increases (GI bleed → blood digested as protein → urea; high protein feeds; steroids → catabolism; catabolic states)
- Kidneys reabsorb more urea (prerenal — avid tubular reabsorption)

Creatinine rises only when GFR falls — it doesn't change with diet or catabolism.

So **BUN:Cr is HIGH** (>20) in:

- Prerenal (reabsorption)
- GI bleed (protein load)
- High protein feeding (Ensure)
- Steroids (catabolism)
- Any catabolic state

BUN:Cr is LOW or NORMAL when creatinine rises **WITHOUT** proportional BUN rise — classic example: **rhabdomyolysis**. Muscle breakdown releases creatinine directly → Cr rises steeply → ratio drops.

1. What ratio tells you

WHAT YOU SEE

Top line: 'BUN:Cr ratio tells you how much urea is in blood relative to creatinine'

WHAT IT ENCODES

BUN:Cr compares two nitrogenous wastes that the kidney handles differently. Urea is freely filtered but PASSIVELY reabsorbed in the proximal tubule and collecting duct following water — so when avid water reabsorption is on (low flow, ADH), urea follows and BUN climbs. Creatinine is filtered and essentially NOT reabsorbed, making it a cleaner GFR marker. The ratio therefore separates 'flow/volume' problems (urea-dominant) from true parenchymal GFR loss (creatinine-dominant).

BOARD TRAP

A rising BUN is a reliable standalone marker of falling GFR. WRONG — BUN is confounded by protein load/catabolism (GI bleed, steroids, feeds) and by avid reabsorption (low flow), all independent of GFR; only creatinine tracks GFR.

2. Normal ratio

WHAT YOU SEE

'Normal = 10–15:1'

WHAT IT ENCODES

Normal BUN:Cr is 10–15:1 (some texts say 10–20:1). In intrinsic renal failure such as ATN, both rise proportionally so the ratio stays NORMAL (~10–15). A normal ratio with elevated creatinine points AWAY from prerenal and toward an intrinsic tubular process.

BOARD TRAP

A normal BUN:Cr ratio rules out kidney disease. WRONG — in ATN/intrinsic AKI both BUN and Cr rise together, keeping the ratio normal even with severe AKI. The ratio discriminates prerenal vs intrinsic, it does not screen for the presence of AKI.

3. Prerenal >20:1

WHAT YOU SEE

'Prerenal = >20:1 (kidneys concentrate urea avidly when perfusion is low)'

WHAT IT ENCODES

Prerenal azotemia → ratio >20:1. Low renal perfusion (hypovolemia, CHF, cirrhosis, sepsis early) triggers RAAS + ADH, driving avid proximal Na/water reabsorption; urea passively follows the water back into blood while creatinine does not. Supporting prerenal labs: FENa <1%, urine osmolality >500, urine Na <20. Treatment is restoring perfusion (IV fluids for hypovolemia, diuresis/afterload reduction for cardiorenal).

BOARD TRAP

A BUN:Cr >20:1 is diagnostic of prerenal AKI. WRONG — GI bleed, high-protein/steroid catabolism, and tetracycline also push the ratio >20 with normal kidneys. Always corroborate with urine Na, FENa, and clinical volume status before calling it prerenal.

4. BUN rises header

WHAT YOU SEE

Section 'BUN rises when:'

WHAT IT ENCODES

BUN is a confounded number: it rises by two independent routes — increased urea PRODUCTION (protein/nitrogen load or catabolism) or increased urea REABSORPTION (low flow). Only the reabsorption route reflects true volume/perfusion status. This is why creatinine, not BUN, anchors GFR estimates.

BOARD TRAP

You can read volume status straight off the BUN. WRONG — only the reabsorption route reflects perfusion; a high BUN from production (catabolism, GI bleed, protein feeds) tells you nothing about volume. Confirm with FENa, urine Na, and exam.

5. ↑Urea production

WHAT YOU SEE

Bullet: GI bleed (blood digested as protein), high protein feeds, steroids → catabolism

WHAT IT ENCODES

Production drivers raise BUN with a NORMAL GFR: upper GI bleed (hemoglobin digested → amino acids → urea), high-protein enteral feeds (e.g., Ensure/TPN), corticosteroids and tetracyclines (anti-anabolic/catabolic), thyrotoxicosis, burns, and fever. These inflate BUN:Cr without any drop in perfusion.

BOARD TRAP

A patient on high-dose steroids with a rising BUN must be volume-depleted. WRONG — steroids are catabolic and raise urea production directly; creatinine and FENa stay normal. Look at the creatinine and urine indices, not the BUN alone.

6. ↑Urea reabsorption

WHAT YOU SEE

Bullet: kidneys reabsorb more urea (prerenal) → avid tubular reabsorption

WHAT IT ENCODES

The reabsorption route is the true prerenal mechanism: hypoperfusion → RAAS/ADH activation → enhanced proximal Na+water reabsorption with passive urea backflow → disproportionate BUN rise vs creatinine. This is the only high-ratio cause that responds to fluids/perfusion correction.

BOARD TRAP

All high BUN:Cr ratios will improve with IV fluids. WRONG — only the reabsorption (prerenal) cause is fluid-responsive; a high ratio from GI bleed, steroids, or protein feeds will not normalize with fluids and may signal the bleed itself.

7. Creatinine behavior

WHAT YOU SEE

Line: 'Creatinine rises only when GFR falls — doesn't change with diet or catabolism'

WHAT IT ENCODES

Creatinine is generated at a steady rate from muscle creatine and excreted almost purely by filtration, so it tracks GFR and is immune to diet/catabolism. Caveats: it LAGS GFR by 24–72h in acute injury (a normal early creatinine can hide severe AKI), and low muscle mass (elderly, cachectic, cirrhotic) gives a falsely low creatinine that overstates GFR. Some drugs (trimethoprim, cimetidine) block tubular creatinine secretion and bump creatinine WITHOUT a true GFR change.

BOARD TRAP

Creatinine is a perfect, instantaneous GFR marker. WRONG — it lags acute GFR loss by days and is biased by muscle mass and by secretion-blocking drugs (trimethoprim, cimetidine). A 'normal' creatinine early in AKI does not equal a normal GFR.

8. BUN:Cr HIGH list

WHAT YOU SEE

'So BUN:Cr is HIGH (>20)' header

WHAT IT ENCODES

Five classic high-ratio (>20:1) causes: (1) prerenal hypoperfusion (reabsorption), plus four production/load causes — (2) GI bleed, (3) high-protein feeds, (4) steroids/catabolism, (5) tetracyclines. Only #1 is a volume/perfusion problem.

BOARD TRAP

A ratio >20:1 on the panel equals prerenal AKI. WRONG — four of the five high-ratio causes (GI bleed, protein feeds, steroids, tetracyclines) are production-driven with normal kidneys and normal FENa; only hypoperfusion is true prerenal physiology.

9. Prerenal (reabsorption)

WHAT YOU SEE

Bullet 'Prerenal (reabsorption)'

WHAT IT ENCODES

Prerenal is the prototype high-ratio state caused by avid tubular reabsorption — confirm with FENa <1%, urine osm >500, urine Na <20, and bland urine sediment (hyaline casts at most). It is fully reversible if perfusion is restored before tubular ischemia converts it to ATN.

BOARD TRAP

Every prerenal AKI shows FENa <1%. WRONG — prerenal on a background of diuretics or CKD can have FENa >1%; use FEUrea (<35% = prerenal) when the patient is on diuretics.

10. GI bleed / protein load

WHAT YOU SEE

Bullet 'GI bleed (protein load)'

WHAT IT ENCODES

An UPPER GI bleed delivers a large protein (hemoglobin) load to the gut; bacterial breakdown → absorbed amino acids → hepatic urea synthesis → BUN spikes while creatinine stays flat → ratio often >30–40. A high BUN:Cr in a patient with melena/coffee-ground emesis is a clue to GI bleeding itself.

BOARD TRAP

A high BUN:Cr from a GI bleed means the patient is simply volume-depleted. WRONG (or only half-right) — the dominant mechanism is the digested-protein urea load, not pure prerenal physiology; the ratio can be very high even before significant hypovolemia. Treat the bleed and assess volume separately.

11. High protein / steroids

WHAT YOU SEE

Bullets 'High protein feeds (Ensure)' and 'Steroids (catabolism)'

WHAT IT ENCODES

High-protein enteral nutrition (Ensure, TPN) and catabolic states (corticosteroids, sepsis, trauma, GI bleed) raise urea production with a normal GFR → isolated BUN elevation and high ratio. Tetracyclines act similarly via an anti-anabolic effect. These resolve when the load/drug is withdrawn.

BOARD TRAP

An ICU patient on tube feeds and steroids with an isolated BUN bump has worsening renal function. WRONG — catabolism plus a high-protein load raises urea production with a normal GFR; the creatinine and urine indices stay normal. Don't escalate fluids or call it AKI on BUN alone.

12. BUN:Cr LOW / normal

WHAT YOU SEE

Bottom: 'BUN:Cr is LOW or NORMAL when creatinine rises WITHOUT proportional BUN rise — classic = rhabdomyolysis'

WHAT IT ENCODES

Low/normal ratio (<10–15) occurs when creatinine rises out of proportion to BUN OR when urea production is low. Causes: (1) rhabdomyolysis — necrotic muscle dumps creatinine directly, so creatinine outruns BUN; (2) advanced liver disease/malnutrition — the failing liver makes little urea so BUN is low; (3) SIADH and overhydration — dilution; (4) low protein intake. A low ratio in a patient with crush injury, statin/seizure/immobility, and CK in the thousands points to rhabdo with pigment nephropathy.

BOARD TRAP

Rhabdomyolysis produces a HIGH BUN:Cr ratio. WRONG — rhabdo releases creatinine straight from damaged muscle, so creatinine rises faster than BUN and the ratio is LOW or normal. The other low-ratio trap is cirrhosis, where a poorly functioning liver underproduces urea and masks azotemia.